

IMPROVING ORAL READING FLUENCY ASSESSMENT THROUGH SUB-SEQUENCE MATCHING OF ACOUSTIC WORD EMBEDDINGS

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ABSTRACT

Oral reading fluency assessment is a process where a student reads a passage aloud and is scored against words read correctly by a human listener. Current automatic reading fluency systems match these words read using speech recognition models trained with clean speech data from native adult speakers. This mismatch in training and deployment, compounded by numerous background noises from the classroom, means that student speech is often not correctly recognized. This paper describes a deep learning model that employs text-to-speech and contrastive learning to create acoustic word embeddings of student speech. This embedding is trained with unlabeled data of students reading known passages. Our model then uses sub-sequence matching in the acoustic embedding space to estimate words read correctly per minute, a common criterion in oral reading fluency. Our model's words read correctly per minute is significantly closer to human listeners compared to systems that use automatic speech recognition only, reducing error of words correct per minute from 15.1 to 8.4, on average.

Index Terms— Reading Fluency, Deep Learning, Subsequence Matching, Acoustic Word Embedding

1. INTRODUCTION

Poor reading ability in early grades is a strong predictor of poor performance in later grades [1]. Therefore, assessment of reading ability through oral reading fluency (ORF) is critical in early education. A common and practical metric that is designed to evaluate students' oral reading skills is words read correctly per minute (WCPM), typically assessed in one-on-one sessions with a student and a human listener [2, 3, 4]. However, assessment of WCPM using human listeners is time consuming which has led to great interest in using automatic speech recognition (ASR) for automation of the assessment of ORF [5, 6]. Previous researchers that attempted to use automatic speech recognition for language learning often reported unreliable results [7]. However, Neri et al. [8] reported that many of these reported errors could be mitigated with proper feedback when the ASR was not correct—but still required manual review and listening. Since then, modern day speech recognition has achieved many milestones in terms of recognizing speech with different accents and noise. However, recognizing student speech in the classroom still remains challenging because many students exhibit unique pronunciations and phrasing when reading aloud [9]. Variances like sudden pace changes, mumbling, or subdued speech often pose issues for conventional speech recognition tools. Recent research has investigated ASR for nonstandard speakers, but is not directly applicable to oral reading fluency [10].

To mitigate these influences, many previous works adopt “passage-specific” methods in assessing fluency such as models

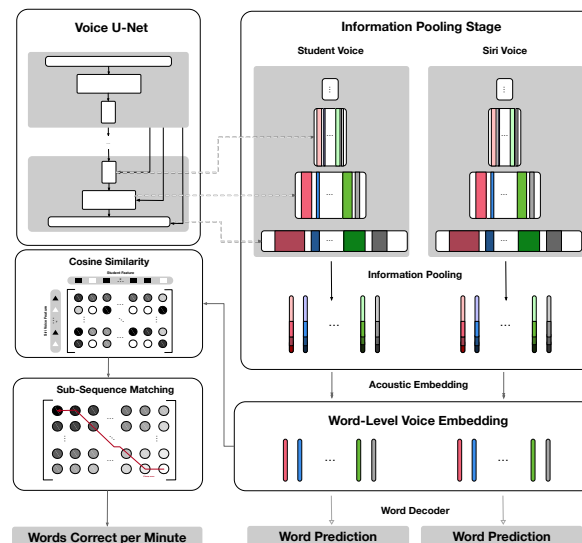


Fig. 1. Overview of the Neural Network

of pronunciation [11, 12, 13, 14] or measurement of prosody as a surrogate for ORF [15, 16]. These methods struggle to adapt to passages that contain a diverse range of words, potentially hindering the accuracy of fluency estimation [17]. In this work, we design a novel method for calculating WCPM from speech using contrastive learning, novel word level acoustic embeddings, and modern text-to-speech (i.e., Apple Siri) to achieve state of the art performance in automated ORF assessment. After finding proximity of words in our acoustic word embedding space, we use sub-sequence matching to estimate the WCPM for a student in a given passage. We find that our method is comparable to the manual ratings of a human listener, differing by 8.7 words per passage, on average. This is a reduction in error of 50% compared to using ASR only methods.

2. METHODOLOGY

2.1. Dataset Description

To evaluate our matching approach along with a commercial speech recognizer, we compare the predicted word matches to the actual word matches as assessed by a linguistic annotator from a previously collected dataset [18] (2014-2020). This dataset contains human scoring of “words correct” in passages from approximately 650 second-grade through fourth-grade students on 319 passages. We also filter the dataset to responses in which students respond verbally

and continue to read for at least ten words.

Dataset Summary: Recordings 14,339 / 13,029 (Total / Filtered), Students: 662 / 653, Passages: 319, Unique Words: 1,674. Students are allowed to continue reading until one minute has passed. Audio files are in PCM WAVE format at 16 kHz sampling rate. Labels incorporate Student ID, Passage ID, and human-rated WCPM, with each Passage ID corresponding to a specific text. Students' WCPM was typically between 50 and 150 words, which is expected for second- through fourth-grade levels [3].

2.2. Data Preparation

The dataset employed does not contain specific information on which words were read correctly per student, only the final WCPM. As such, our analysis is limited to the audio files and the designated passage text for students. The specifics of the students' utterances remain unknown. Consequently, we forwarded this data to Apple's ASR, providing transcripts of the words spoken and their specific segments. In this context, a segment delineates the beginning and end of a word. Notably, if a student articulates swiftly, two words might fit into a single segment. We then juxtaposed the transcriptions of student recordings with the reference text and compared them through the longest common sub-sequence (LCSS) algorithm. This comparison was based on the word stems presentation of transcriptions and the reference text. We determined the WCPM using the start time of the first aligned word and the end time of the last. This methodology serves as our foundational benchmark.

Next, we used the 2023 Siri voice, an on-device feature on iOS devices, which uses Apple's deep mixture density networks to derive a reference for voice features [19]. While we cannot directly access the deep mixture density networks, we can harness the voices it produces. We handpicked four variations to create reference voices for our passages from the vast pool of voice actors associated with this Siri version. Finally, we fed the generated voice back into Apple's ASR to extract word and segment details. We then employed the LCSS algorithm to align each Siri Voice individually, as even the segments from Siri's voice can encompass multiple words. Additionally, it's noteworthy that different Siri voices have distinct merged segments.

2.3. Neural Network Architecture

We design a series of neural networks to transform input audio passages into embeddings of the student voice for each word. The neural networks can be separated into four individual parts: the Voice-based U-Net, a dynamic Region of interest pooling network, a feed-forward network for generating embeddings, and a feed-forward network for decoding the voice embeddings back to words, as shown in Fig. 1. An open source implementation of the model is available ¹.

2.3.1. Voice U-Net

U-Net [20] is a well known fully convolutional auto-encoder. It is augmented with skip connections that reintroduce fine details that might get lost during encoding, a strategy prevalently used in computer vision for image-to-image transformations. Leveraging the U-Net model, we can develop a hierarchical multi-resolution of features, which is a common practice in deep learning. Our approach is similar to Wave-U-Net [21] used for audio enhancement [22] and source separation [23], but appropriately adapted for use at the word

level. That is, our U-Net architecture uses input from the Mel-spectrogram and does not use reconstruction loss for pre-training. Rather, our U-Net architecture is trained in tandem with the other networks. We also make task specific modifications to the U-Net implementation, departing from the standard architecture. We choose average pooling for downsampling, and in the upsampling stage, we adopt transposed convolution. Furthermore, each weighting block leans on principles from ResNet [24], weaving in a linear identity map throughout the network. In all non-linear branches, we position a dropout layer before the weighting layer, which is helpful to mitigate overfitting.

2.3.2. Information Pooling

Information pooling employs statistical pooling on the Region of Interest (RoI) in each audio segment, leveraging the Apple ASR produced data. The associated RoI is extracted from all feature maps during the U-Net decoder phase for every segment. Subsequently, average pooling is applied to this RoI and all the pooled feature maps are concatenated. Our approach closely aligns with the principles of Atrous Spatial Pyramid Pooling (ASPP) [25]. However, while ASPP generally employs dilated convolutions, given the U-Net structure, our method simply pools from various feature map sizes. This avoids the potential aliasing artifacts commonly associated with dilated convolutions in residual networks [26]. These multi-resolution features can let the network extract words that have vastly different lengths, as shown in Fig.1 (top right). Once pooling is applied, our methods can be thought of as acoustic word embeddings [27, 28], but do not rely on pre-trained recognizers. Instead, we use a mixture of real and generated speech, with contrastive learning, to urge our acoustic word embeddings to be relevant in the application of automated ORF assessment.

2.3.3. Acoustic Word Embedding Network and Decoding Network

The acoustic word embedding draws a vector from the information pooling layers and then employs a residual feed forward network to reduce its feature dimensionality. The output of this embedding is used for the contrastive loss, the specifics of which will be elaborated on later. The embedding decoding network strives to extract the stem of the word from the embedded features. The ultimate output correlates with the total number of word stems identified in the dataset.

2.4. Neural Network Training Procedure

During the training phase, we matched each student's response to a specific passage with Siri Voice actor's rendition of that passage. The following procedure was employed (as shown in Fig. 1): First, for both the student recordings and Siri's Voice, a Mel-Spectrogram was produced. The LCSS was subsequently applied to these paired datasets, from which we discerned and tagged the relevant segments based on matched segments. Any segment that is unmatched was discarded. Second, the student recordings and their Apple Siri counterpart were concurrently input into the voice U-Net. The ensuing outputs were channeled to the information pooling network, which leveraged the data from relevant segments to extract vectors for each word. Third, these pooled vectors were subsequently directed to the acoustic word embedding network. We applied a triplet loss [29] on the network's output using cosine distance. All potential pairs were calculated from the resulting embedding vectors between the

¹<https://github.com/YihaoJW/ORFLCSS>

student's voice and Siri's. Embeddings from relevant segments associated with identical word stems were classified as positive; otherwise, they were deemed negative (See Fig. 2). The total triplet loss is calculated as follows.

$$\begin{aligned}\bar{d}_+ &= \frac{1}{|P|} \sum_{i \in P} d_{+i} & \bar{d}_- &= \frac{1}{|N|} \sum_{j \in N} d_{-j} \\ d_{\text{edge}+} &= \min_{i \in P} d_{+i} & d_{\text{edge}-} &= \max_{j \in N} d_{-j} \\ L_{T\text{-total}} &= \alpha + \frac{1}{2} (-\bar{d}_+ + \bar{d}_- - d_{\text{edge}+} + d_{\text{edge}-})\end{aligned}\quad (1)$$

Finally, the embeddings were fed into the decoding network to decode the word's stem. This was succeeded by cross entropy loss (CE). We take the average between students' and Siris' CE loss as follows:

$$L_{\text{CE-total}} = \frac{1}{2} (L_{\text{CE-student}} + L_{\text{CE-siri}}) \quad (2)$$

While training the network, the two losses are aggregated using uncertainty to weigh losses [30]. In equation (3), σ_1 and σ_2 are trainable parameters.

$$L = \frac{1}{2\sigma_1^2} L_{T\text{-total}} + \frac{1}{\sigma_2^2} L_{\text{CE-total}} + \log(\sigma_1 \sigma_2) \quad (3)$$

2.5. Sub-Sequence Matching using Embedding of Word

After completing the neural network training, we can process all segments from both the student recordings and each Siri actor's voice to obtain the voice embedding for each segment. We introduce a new matching algorithm derived from the longest common sub-sequence concept tailored to our embedded vectors. Instead of striving to match the longest possible sequence, our objective is to attain the global maximum of sequence fitness score by giving acoustic word embedding as inputs. The fitness score is determined by the cumulative cosine similarity of all matched segments that exceed a threshold α . The state transition function is given in equation (4). $T(i, j)$ represents the fitness score at the position i, j . The function $s(i, j)$ denotes the cosine similarity between the student recording segment i and Siri Voice segment j if this similarity is greater than α otherwise $s(i, j) = -\infty$:

$$\begin{aligned}T(i, 0) &= 0, \quad T(0, j) = 0, \quad \forall i, j \\ T(i, j) &= \max \begin{cases} T(i-1, j) \\ T(i, j-1) \\ T(i-1, j-1) + s(i, j) \end{cases} \quad (4)\end{aligned}$$

Once the highest fitness value is identified within the table, a back-trace can be executed to ascertain all the corresponding matched segments. The algorithm is run to compare all available Siri actors' voices in our dataset and use the result with the highest fitness score as the final result.

3. RESULTS

The dataset is divided into training and validation sets roughly in 80% and 20%, each containing an equal number of unique passages. Students in the validation set are distinct from students in the training set, ensuring no overlap. This separation is intentional: while our model is crafted not to be overly dependent on specific passages, we are more focused on eliminating student-specific biases rather than passage dependencies in this cross-validation.

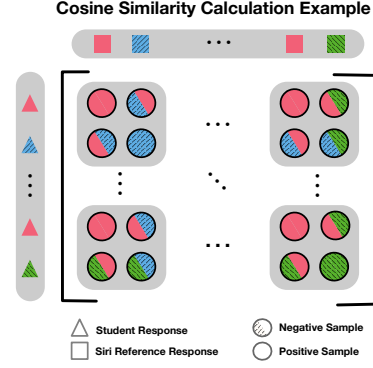


Fig. 2. Example of Cosine Similarity Calculation

During the network training, we employed augmentation by pairing each student with various Siri actors' voices. The computations were executed on a NVIDIA DGX SuperPOD 8xA100 system. The training set comprised over 40,000 unique pairs, and the model was trained for more than 250 epochs. Completing the training process required about 8 days per model.

3.1. Comparison of Model Parameterizations

We investigate the model using different sets of parameters in Table 1 which shows the top-1 and top-5 accuracy of word level prediction on the validation data. dp_r is dropout ratio, d_{emb} is voice embedding size N indicates if the input Mel-frequency spectrogram is normalized, and α is the margin in triplet loss.

Exp. ID	dp_r	d_{emb}	N	α	Top-1	Top-5
1	0.0	64		0.4	0.480	0.862
2	0.0	64		0.8	0.600	0.887
3	0.2	32		0.8	0.836	0.916
4	0.2	64		0.8	0.901	0.973
5	0.2	64	✓	0.4	0.839	0.966
6	0.2	32	✓	0.8	0.870	0.971
7	0.2	32	✓	0.4	0.827	0.952
8*	0.2	128	✓	0.8	0.875	0.973
9*	0.2	128		0.8	0.88	0.977

Table 1. Network performance with different parameters. * denotes models chosen for additional matching evaluation.

Based on the displayed results, it's evident that the inclusion of dropout plays a pivotal role in enhancing performance by curbing overfitting. Additionally, having a substantial margin in the contrastive loss (i.e., a larger α) bolsters the classification outcomes on the stem of words. However, normalizing the Mel-spectrogram appears to strip away some essential information from the audio recordings, leading to a decline in performance. To investigate this further, we choose Exp. ID 8 and ID 9 to continue evaluation.

3.2. Comparison of Matching Result

We undertook a comparative analysis of the performance disparities among human raters, our baseline model, and our latest proposed technique. Using the student recordings from the validation set, we calculated the WCPM with our methods and compared it with the

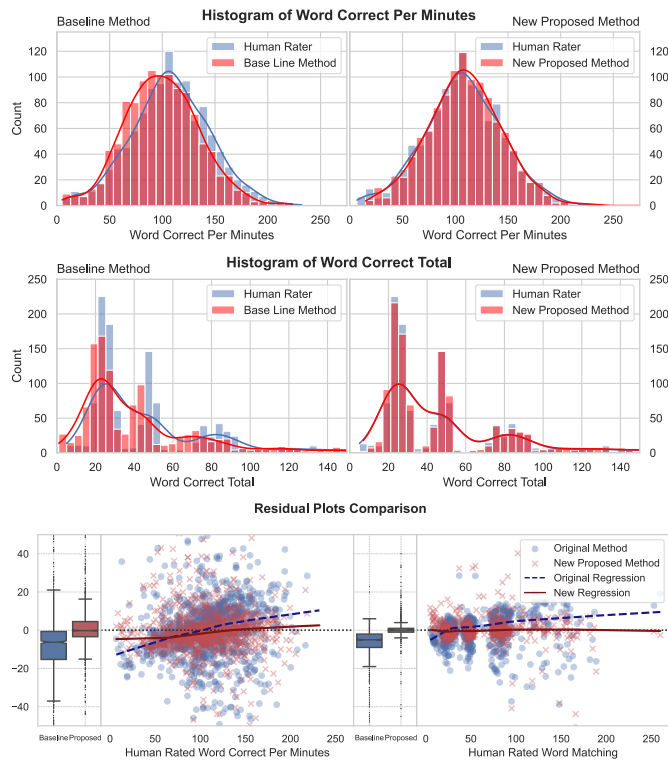


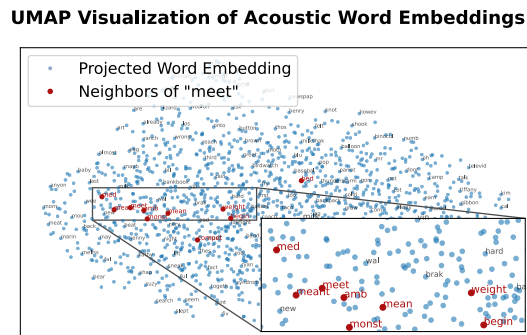
Fig. 3. Comparison between Baseline and Proposed Method

baseline ASR approach. Our baseline method uses ASR and subsequence matching of stemmed words. On the other hand, the new proposed approaches rely on a neural network to generate embeddings. We first assessed the outcomes from models where normalization was either eliminated or retained (Exp. ID 9 or ID 8), in contrast to the baseline method. After analyzing all results from the validation set, we treated each passage as a distinct data point. To gauge the robustness of these results, we applied bootstrapping on the dataset, performing random sampling on each iteration. ‘Match Count Increase’ in Table 2 presents the improvement over baseline, with a 95% Confidence Interval (CI). Our proposed method shows significant improvement, with Mean Absolute Error (MAE) reducing from a baseline of 15.1 to 8.4. Additionally, the MAE for total word matches between the model and human judgments decreased from a baseline of 7.6 to 2.4. Notably, the chosen neural network was the one that exhibited the highest top-5 accuracy (Exp. ID 9 in Table 1). Therefore, subsequent experiments are meant to better characterize this model.

Measurement	Words Match Increase		WCPM MAE	Words MAE
	Mean	Median		
Raw Exp. ID 9	(7.2, 7.8)	(5, 6)	8.4	2.4
Norm Exp. ID 8	(3.6, 4.2)	(2, 3)	13.5	4.5
Baseline	—	—	15.1	7.6

Table 2. Performance Difference between Neural Network

The histograms shown in Fig. 3 (top and middle) reveal a notable bias in the distribution of the baseline method, whereas our method aligns more closely with the human rater’s assessment for both WCPM and words matched per passage. Visually, it is clear

**Fig. 4.** UMAP of Acoustic Word Embeddings

that the proposed approach more closely resembles the expected distribution of WCPM across students. We constructed a residual plot to juxtapose the baseline and proposed methods in relation to the human rater’s assessments. Fig. 3 (bottom) shows that the new method has less residual error compared to the originally proposed method across a range of true WCPM and across words matched per passage.

3.3. Acoustic Word Embedding Visualization

We also visualize the embedding space using UMAP [31] of the acoustic word embeddings from our model, shown in Fig. 4. Given that we employ an embedding decoding network to decipher the embedded vectors back into the stems of words, extracting the average acoustic word embedding from the last layer is feasible. This provides us with visualization of the embedding representation for the word “meet” spoken by a student. The nearest words are shown in the callout of Fig 4, showing that the most common acoustic confusions are words like mean, meant, and ambient.

4. CONCLUSION

In conclusion, our proposed method reduces the difference in human rated versus automated WCPM by 50% compared to using a commercial ASR. The structure is designed to accept a wide range of different passages, over 300 in this experiment, and can achieve good performance without the need for student specific calibration. However a limited number of specific accents were investigated since data collection was limited to a geographical location in the Northwest United States. Sampling student voices in a more diverse region could mitigate these issues since network structure should be invariant to these changes.

Despite this promising performance, there are some technical limitations: Apple ASR can sometimes merge words even if they do not form a common phrase. Additionally, the ASR output can represent equivalent words in varying ways, such as distinguishing between “10” and “ten” or “tenth” and “10th”. Siri Voice only allows the export of low sampling rate and highly compressed voices, which can sound unnatural to the human ear. We believe this is a security feature designed to stop potential abuse of the Siri Voice system, but it might also cause performance issues in research applications. Further, in our approach, we only generate pairs of Siri to user in our contrastive learning, not user to user. Finally, our proposed approach relies on commercial Apple ASR to generate start and stop time of words. This reliance could be removed by using a custom word segmentation method.

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